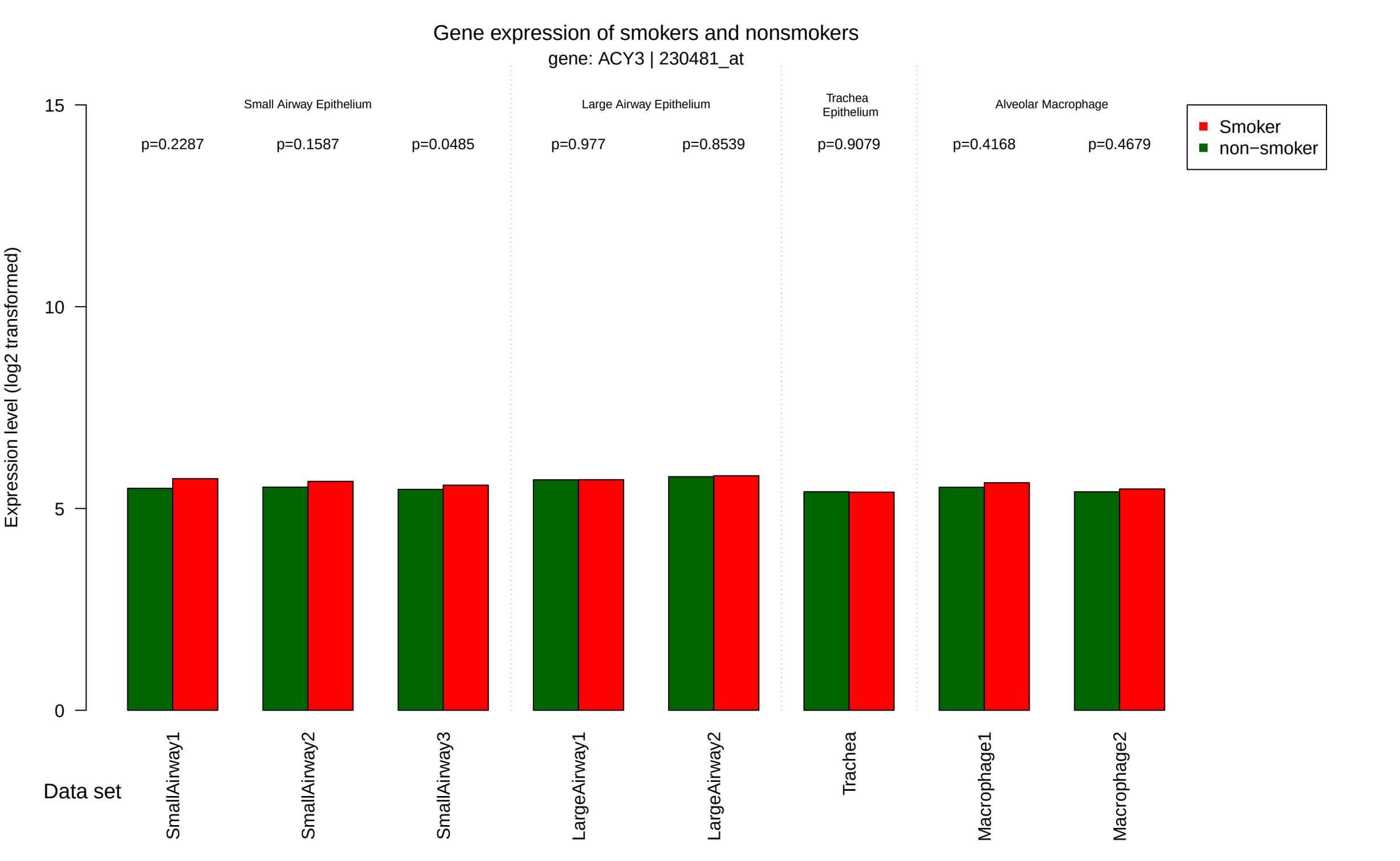
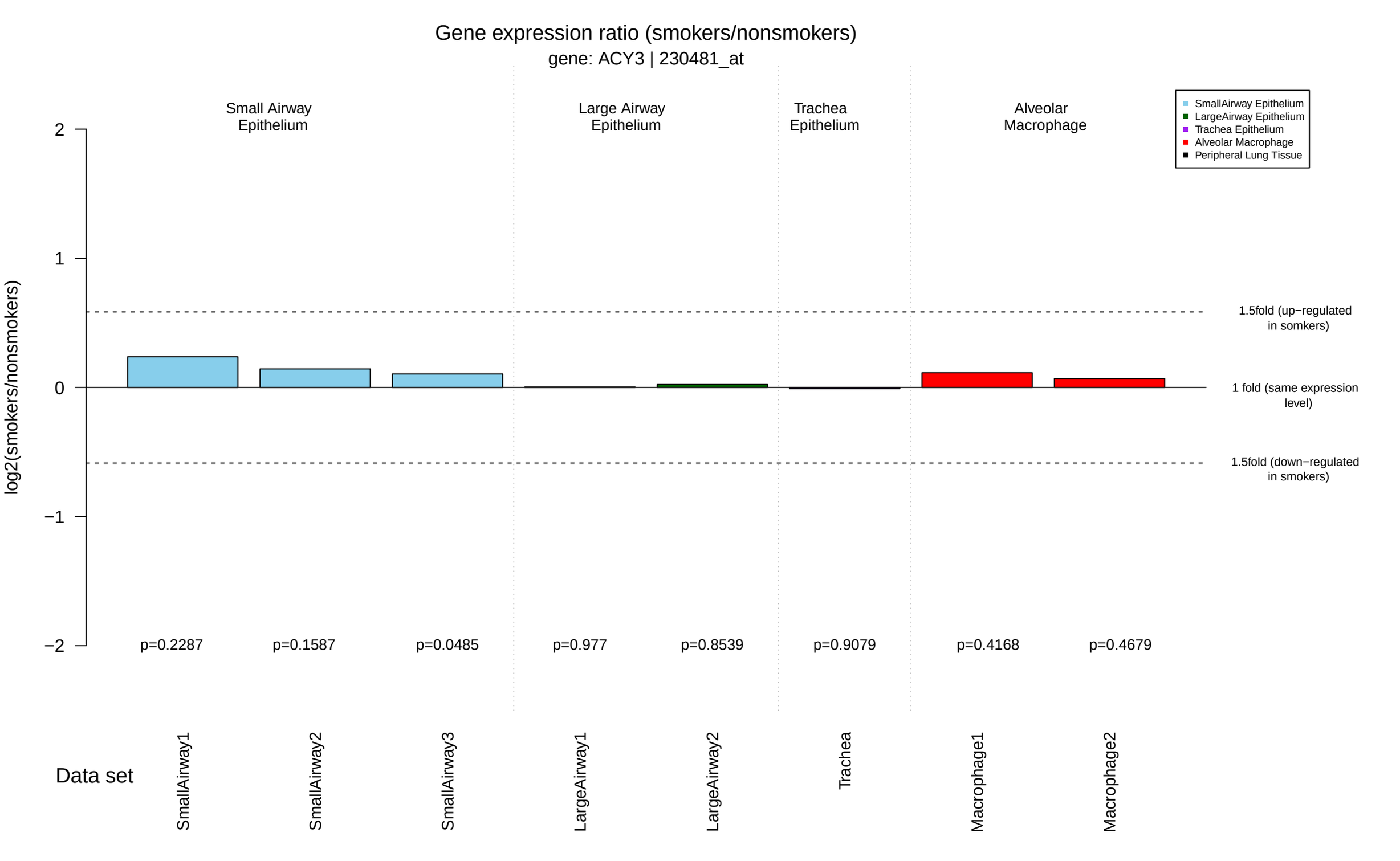
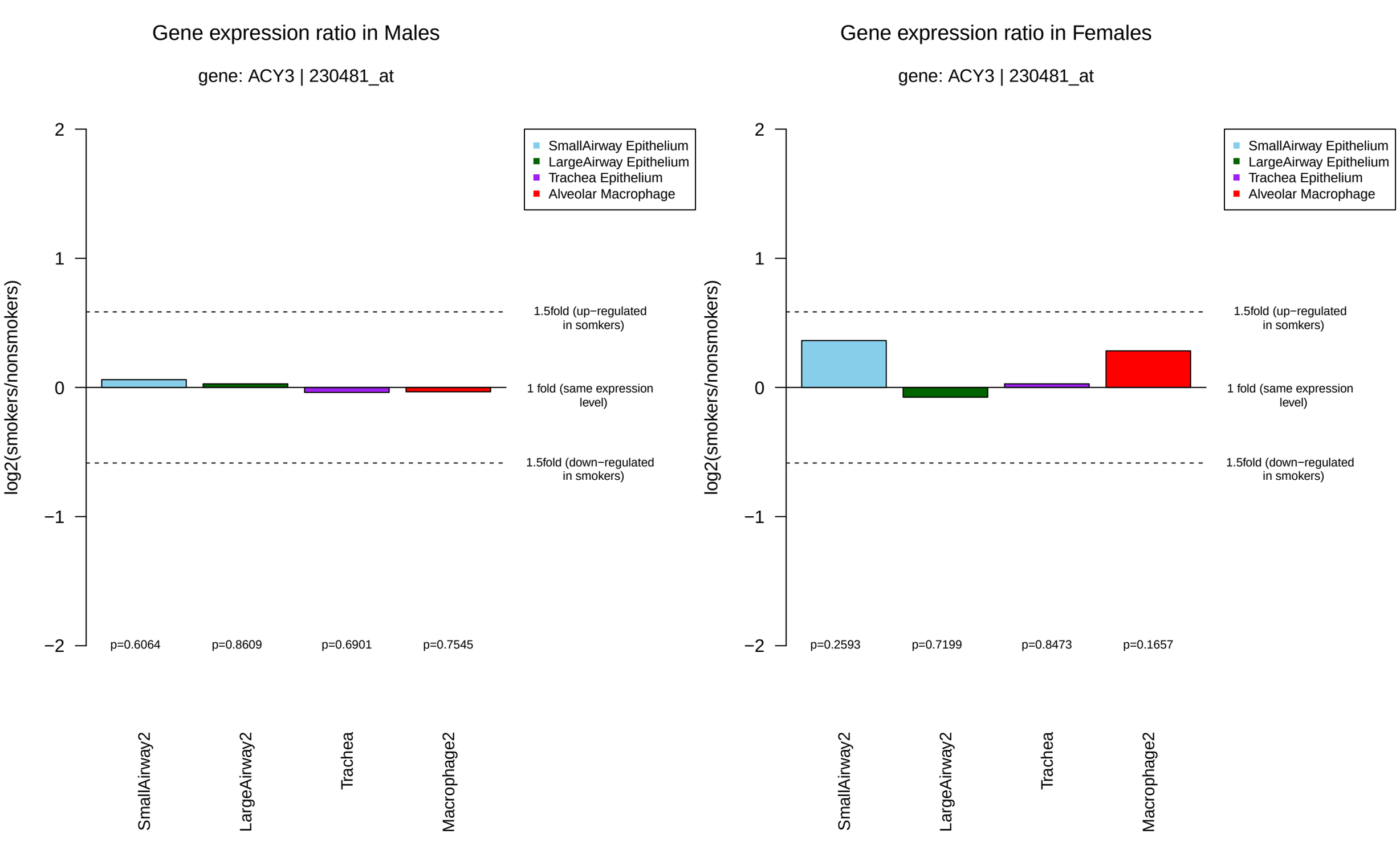
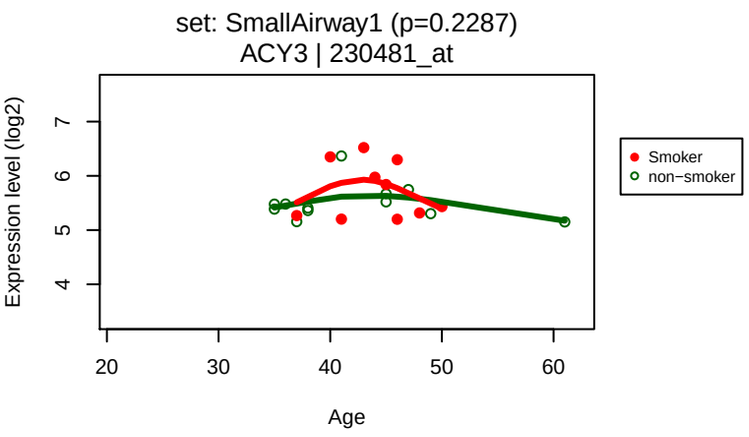
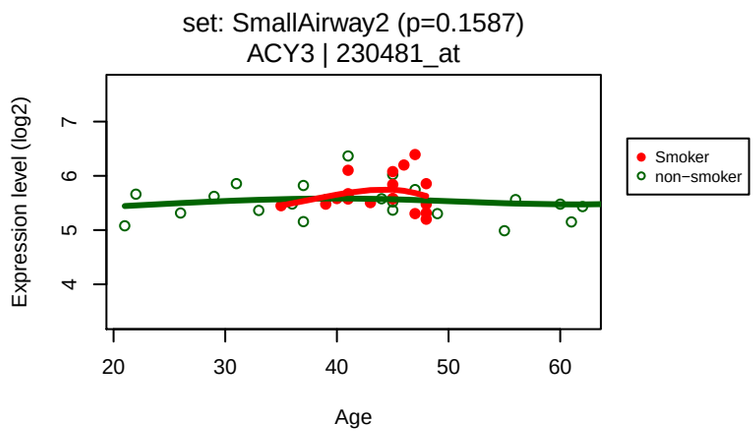
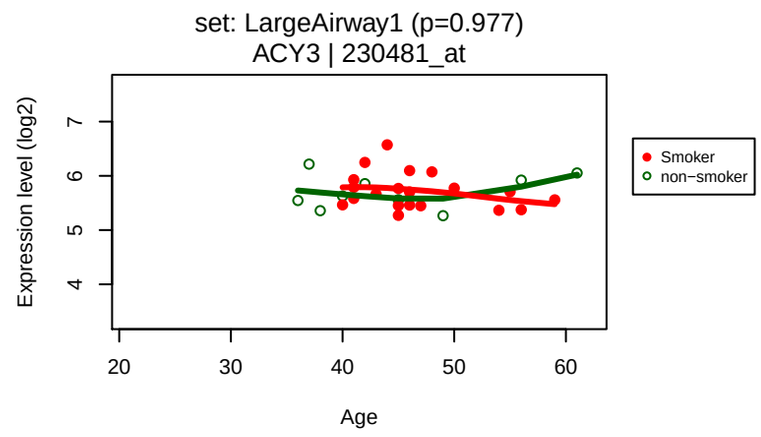

Gene expression of smokers and nonsmokers
gene: ACY3 | 230481_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2287
p=0.1587
p=0.0485
p=0.977
p=0.8539
p=0.9079
p=0.4168
p=0.4679
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

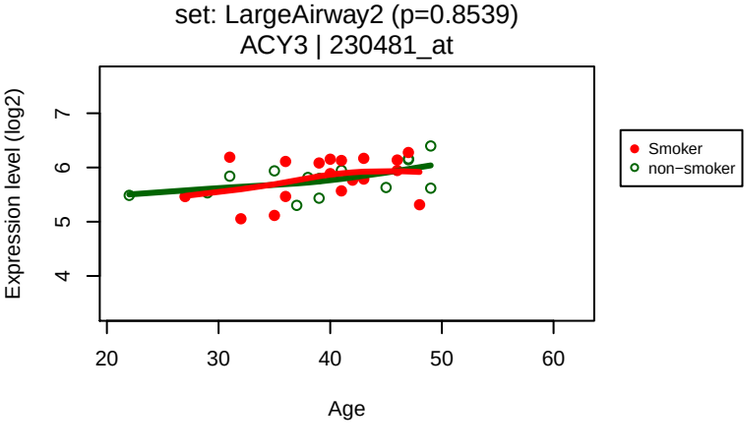

Gene expression ratio (smokers/nonsmokers)
gene: ACY3 | 230481_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2287
p=0.1587
p=0.0485
p=0.977
p=0.8539
p=0.9079
p=0.4168
p=0.4679
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

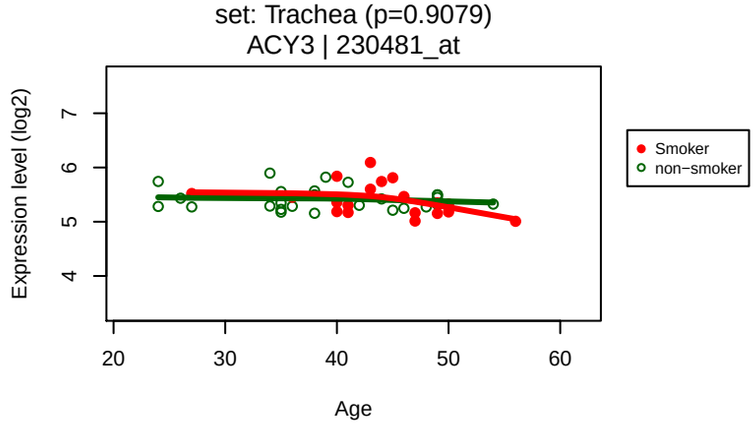

Gene expression ratio in Males
gene: ACY3 | 230481_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6064
p=0.8609
p=0.6901
p=0.7545
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACY3 | 230481_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2593
p=0.7199
p=0.8473
p=0.1657
SmallAirway2
LargeAirway2
Trachea
Macrophage2

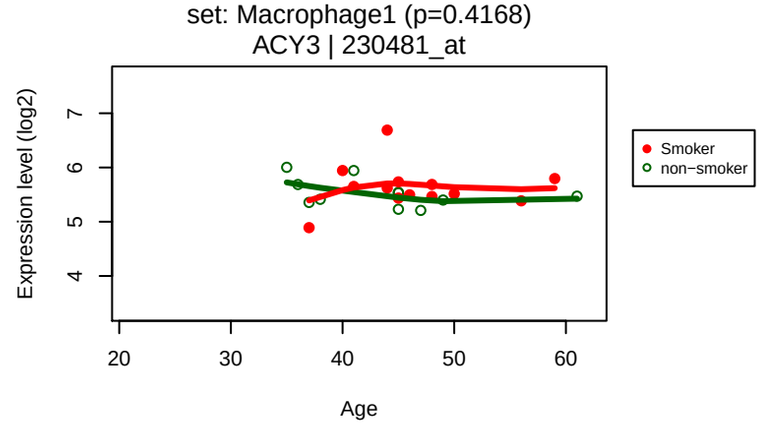

set: SmallAirway1 (p=0.2287)
ACY3 | 230481_at
Expression level (log2)
Age
Smoker
non−smoker

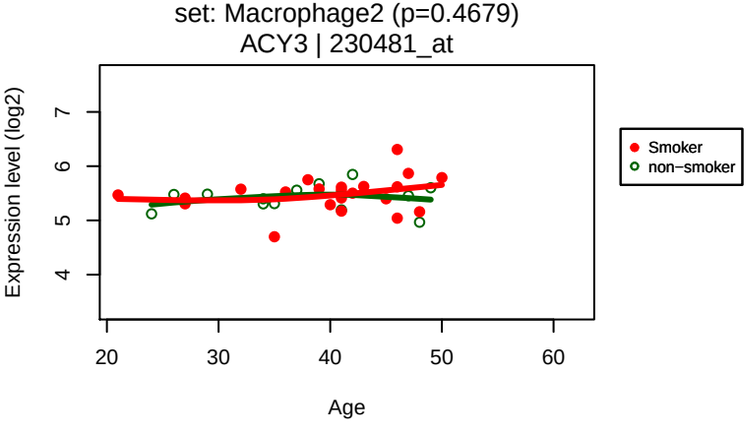

set: SmallAirway2 (p=0.1587)
ACY3 | 230481_at
Expression level (log2)
Age
Smoker
non−smoker

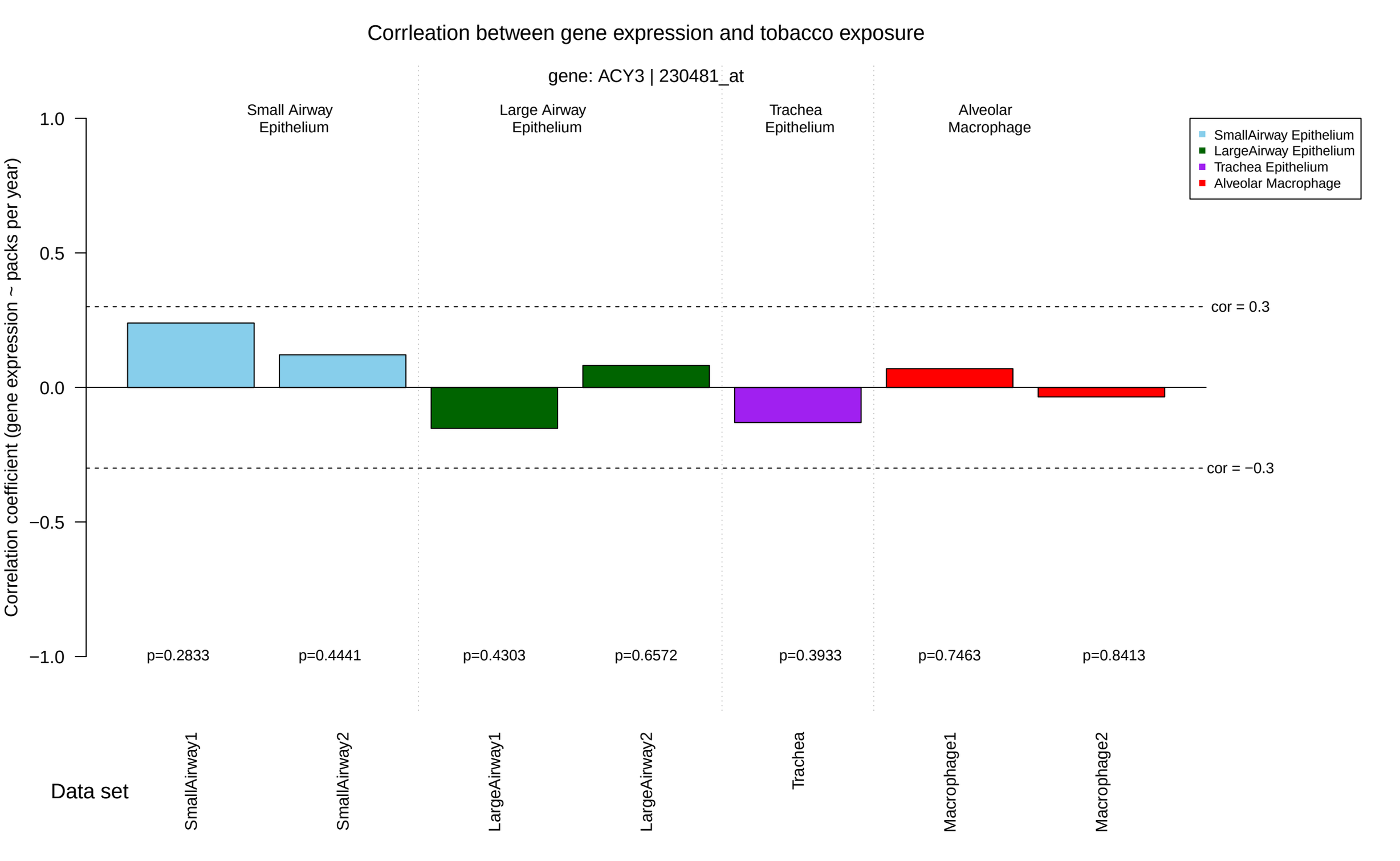

set: LargeAirway1 (p=0.977)
ACY3 | 230481_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8539)
ACY3 | 230481_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9079)
ACY3 | 230481_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4168)
ACY3 | 230481_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4679)
ACY3 | 230481_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACY3 | 230481_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2833
p=0.4441
p=0.4303
p=0.6572
p=0.3933
p=0.7463
p=0.8413
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2